

SLIP – 1

Section I: Data Structures [15 Marks]

Q 1) Write a C program to create random n elements and sort it in ascending order by using insertion sort.

Answer:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int a[100], n, i, j, key;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
        return 0;
    }

    for(i = 0; i < n; i++)
        a[i] = rand() % 100;

    printf("Original array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    for(i = 1; i < n; i++)
    {
        key = a[i];
        j = i - 1;

        while(j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j--;
        }
        a[j + 1] = key;
    }

    printf("\nSorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
}
```

```
    return 0;
}
```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database(primary keys are underlined).

Room (roomno,room_name,room_type,charges)

Guest(Guestcode, Gname,city)

The relationship is as follows: Room-Guest: one-to-one. room_type can have values as either 'AC' or 'NonAC'.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Guest(
    Guestcode INT PRIMARY KEY,
    Gname VARCHAR(50),    city
    VARCHAR(50)
);

CREATE TABLE Room(    roomno
    INT PRIMARY KEY,
    room_name VARCHAR(50),
    room_type VARCHAR(10)
        CHECK(room_type IN ('AC','NonAC')),
    charges NUMERIC(10,2),
    Guestcode INT UNIQUE REFERENCES Guest(Guestcode)
);
```

```
INSERT INTO Guest VALUES
(1,'Amit','Pune'),
(2,'Neha','Mumbai'), (3,'Raj','Nashik');
```

```
INSERT INTO Room VALUES
(101,'Deluxe','AC',12000,1),
(102,'Standard','NonAC',6000,2),
(103,'Suite','NonAC',15000,3),
(104,'Basic','AC',4000,NULL);
```

Q 2(B)

B. Execute the following queries in PostgreSQL (any 3) [6 marks]

i. List the details of the rooms having charges between 5000 and 10000.

Answer:

```
SELECT * FROM Room
WHERE charges BETWEEN 5000 AND 10000;
```

ii. List the names of the guests in the sorted order by city name.

Answer:

```
SELECT Gname FROM Guest ORDER
BY city;
```

iii. List the minimum charges of a room.

Answer:

```
SELECT MIN(charges) FROM Room;
```

iv. Increase the charges of all AC rooms by 15%.

Answer:

```
UPDATE
```

Q 2(C)

C. Write a query to list the name of the guest to whom the room with highest charges is allotted. [4 marks]

Answer:

```
SELECT G.Gname
FROM Guest G
JOIN Room R ON G.Guestcode = R.Guestcode
WHERE R.charges = (SELECT MAX(charges) FROM Room);
```

Q 3) Viva [5 marks]